Experiment 07: Deploying and Managing Containerized Applications using Kubernetes

Experiment No.	07
Mapped CO	CO4
Aim	To deploy and manage containerized applications using Kubernetes.
Tools / Software	Minikube or kind local cluster; kubectl; Docker Engine; A container image from earlier Docker experiment

Theory

Kubernetes is a container orchestration platform used to deploy, expose, scale and manage containerized applications across a cluster. It automates many operational tasks such as scheduling containers, maintaining the desired number of running instances and restarting failed workloads.

The smallest deployable unit in Kubernetes is a Pod. A Deployment manages Pods through ReplicaSets and keeps the desired number of replicas running. A Service provides a stable network endpoint for accessing the Pods, even when Pods are recreated or scaled.

Kubernetes follows a declarative model. The user writes YAML manifest files describing the desired state, such as number of replicas, container image, exposed port and service type. Kubernetes controllers continuously compare actual state with desired state and reconcile the cluster accordingly. This makes applications more reliable, scalable and manageable.

Procedure

1. Install Docker, kubectl and Minikube or kind on the local machine.
2. Start the Kubernetes cluster using minikube start or kind create cluster.
3. Verify the cluster using kubectl get nodes.
4. Prepare or pull a container image created in the Docker experiment, for example <user>/myapp:1.0.
5. Create a deployment.yaml file specifying application name, container image, container port and replica count.
6. Create a service.yaml file of type NodePort to expose the application outside the cluster.
7. Apply both YAML files using kubectl apply -f deployment.yaml -f service.yaml.
8. Check Pods, Deployments and Services using kubectl get pods, kubectl get deployments and kubectl get svc.
9. Access the application using minikube service myapp-svc --url or through the NodePort URL.
10. Scale the deployment using kubectl scale deployment myapp --replicas=4 and observe the new Pods.
11. Delete one Pod manually and verify Kubernetes self-healing by checking that a replacement Pod is created.
12. Clean up resources using kubectl delete -f deployment.yaml -f service.yaml and stop the local cluster.

Sample Code / Commands

deployment.yaml

```
apiVersion: apps/v1
kind: Deployment
metadata:
  name: myapp
spec:
  replicas: 2
  selector:
    matchLabels:
      app: myapp
  template:
    metadata:
      labels:
        app: myapp
```

```
    spec:
      containers:
      - name: myapp
        image: <user>/myapp:1.0
        ports:
        - containerPort: 5000
```

service.yaml

```
apiVersion: v1
kind: Service
metadata:
  name: myapp-svc
spec:
  type: NodePort
  selector:
    app: myapp
  ports:
  - port: 80
    targetPort: 5000
```

Kubernetes commands

```
minikube start
kubectl get nodes
kubectl apply -f deployment.yaml -f service.yaml
kubectl get pods
kubectl get deployments
kubectl get svc
minikube service myapp-svc --url
kubectl scale deployment myapp --replicas=4
kubectl get pods
kubectl delete pod <pod-name>
kubectl get pods
kubectl delete -f deployment.yaml -f service.yaml
minikube stop
```

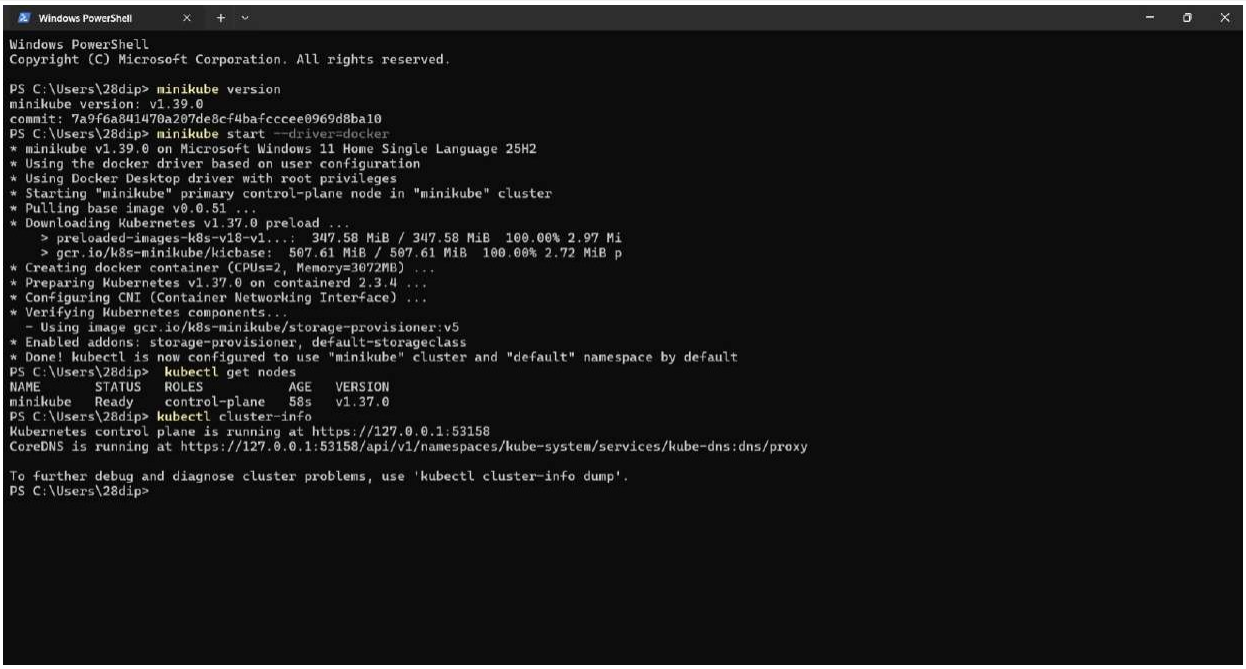
Observation / Experiment Output

Instructions to students: Paste screenshots or copied outputs in the boxes below after completing Kubernetes cluster setup, application deployment, service exposure, scaling and self-healing verification. Keep screenshots clear and readable.

Name	Roll No.	Batch	Date
Vighnesh Pingale	24107108	B2	03-10-2026

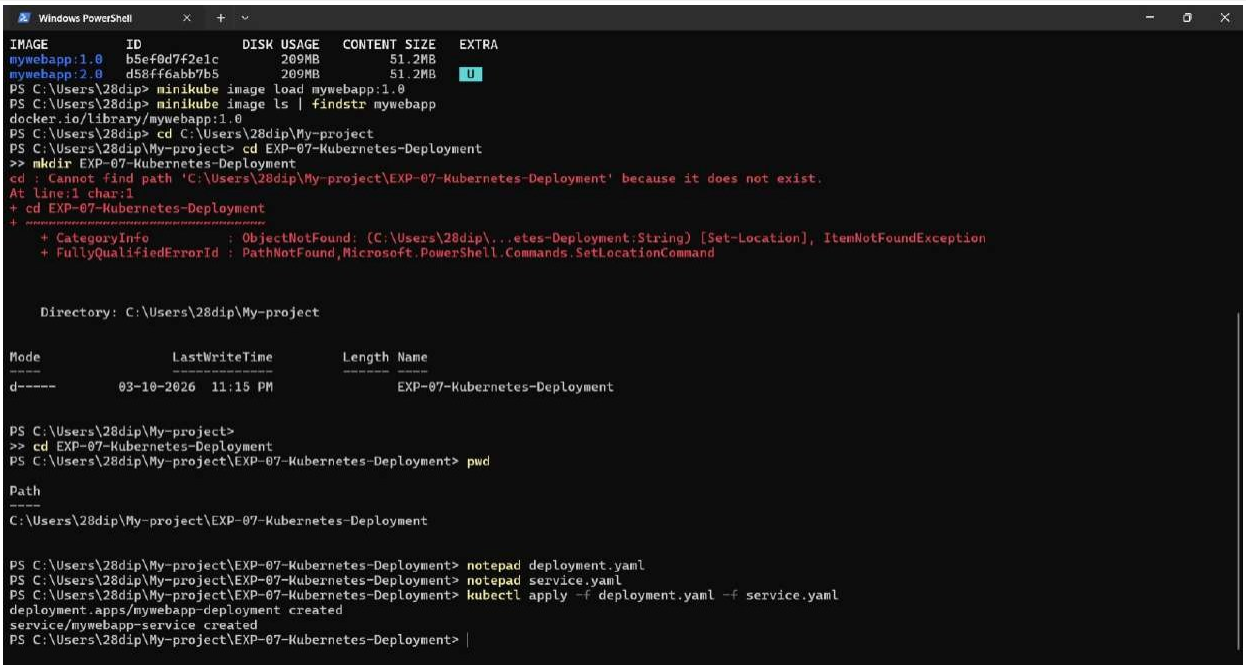
1. Kubernetes Cluster Setup Output

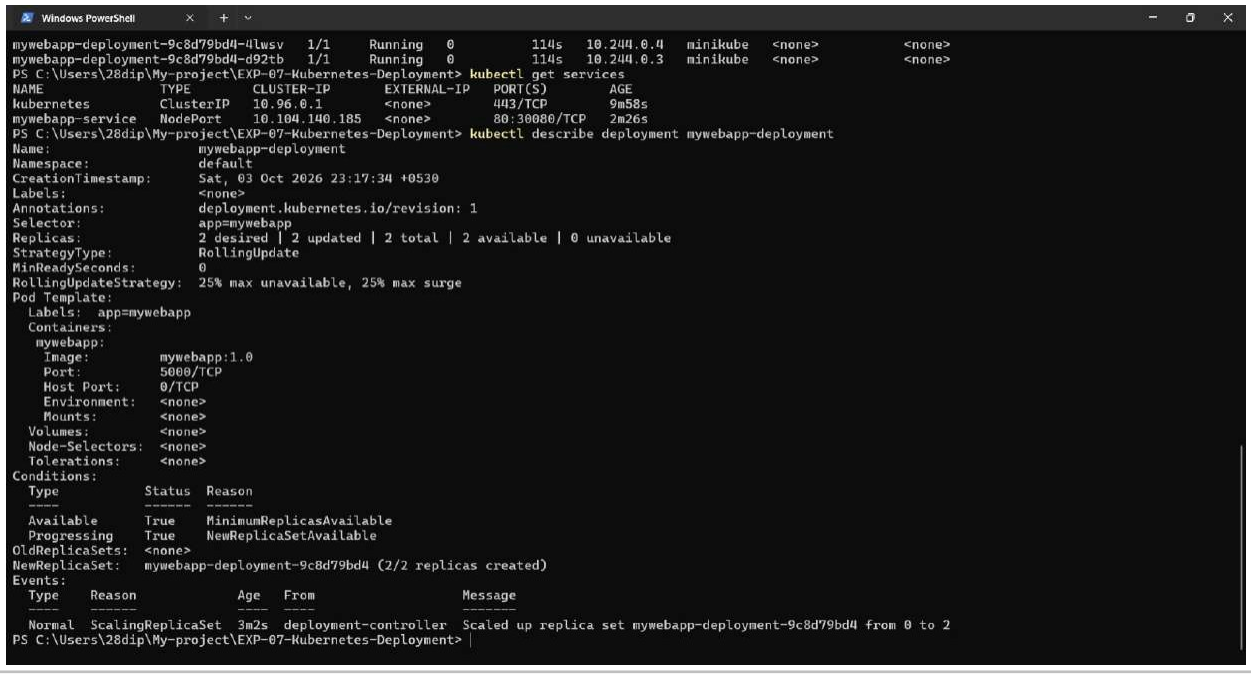
Paste screenshot/output showing minikube/kind start command and kubectl get nodes output.



2. Deployment Manifest Output

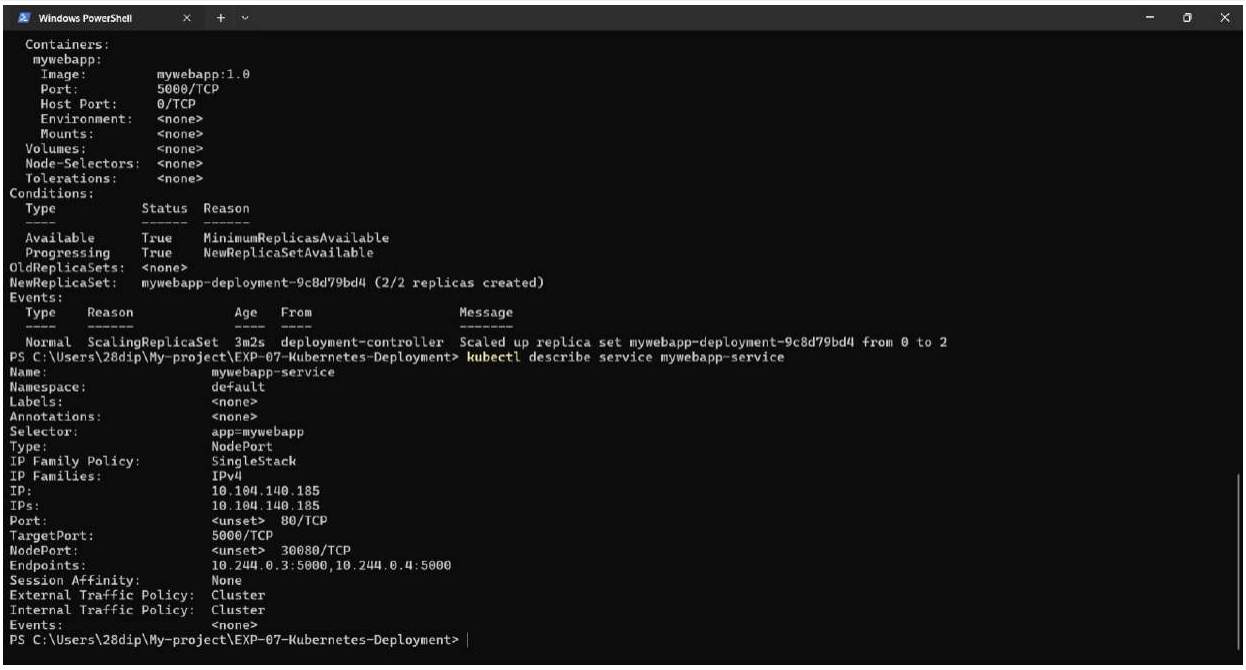
Paste screenshot/output showing deployment.yaml with image name, replicas, labels and container port.





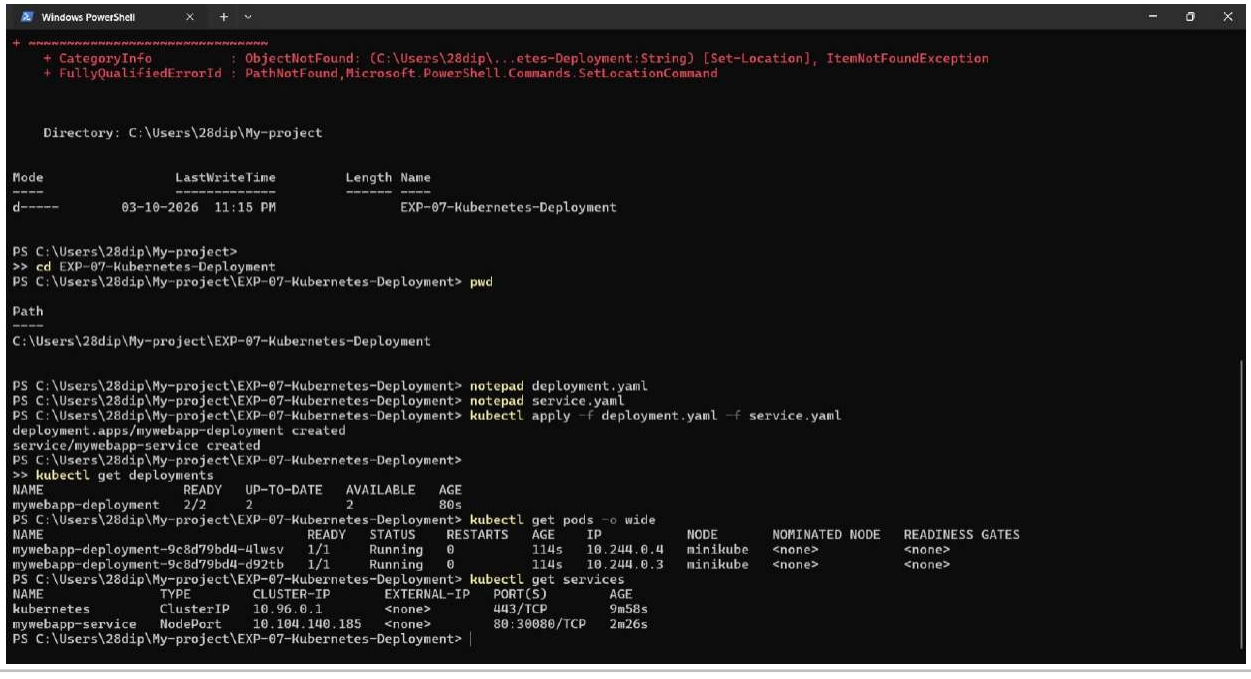
3. Service Manifest Output

Paste screenshot/output showing service.yaml with NodePort/LoadBalancer type, selector and target port.



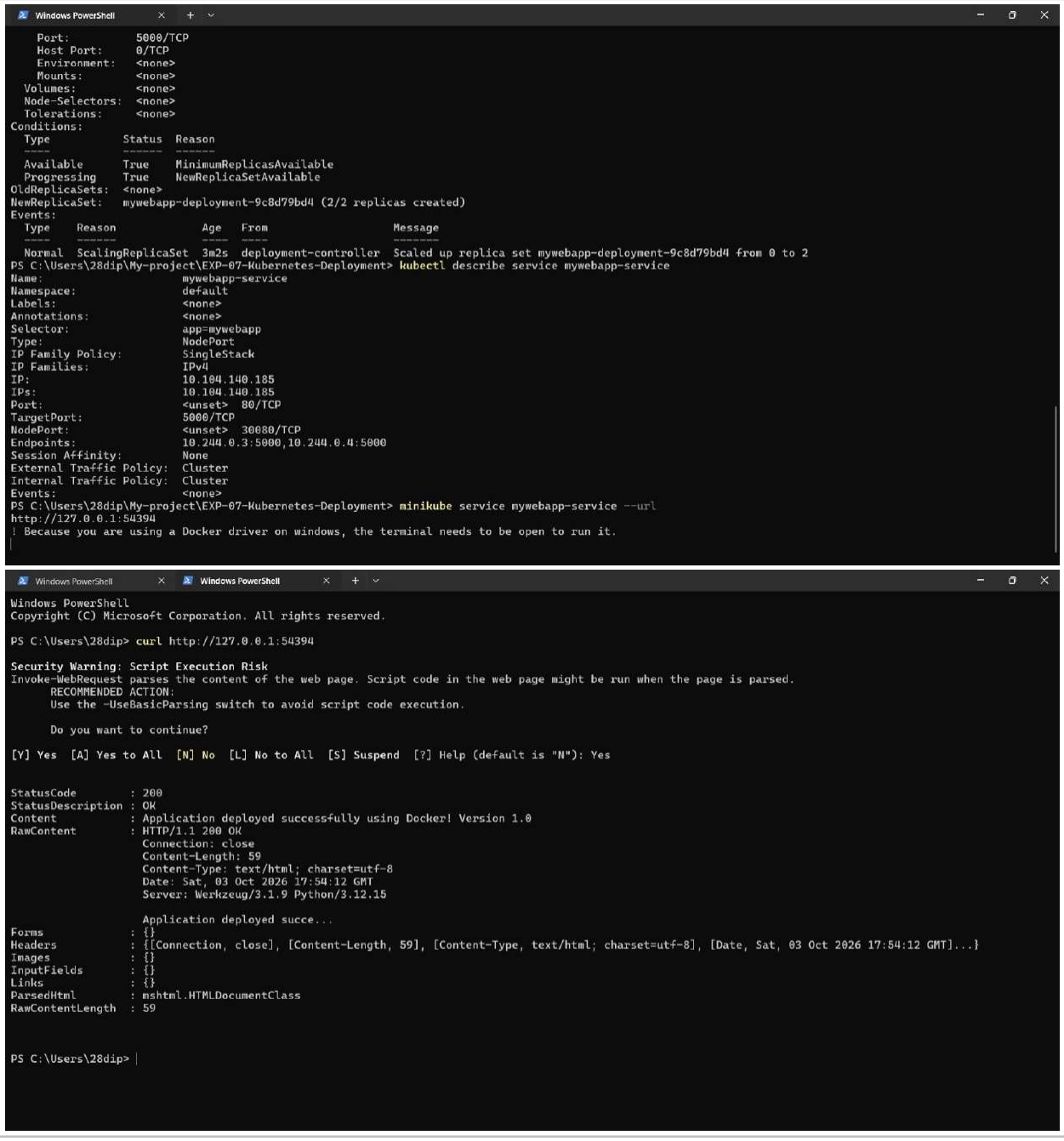
4. Apply and Resource Status Output

Paste screenshot/output showing kubectl apply and kubectl get pods, deployments and services output.



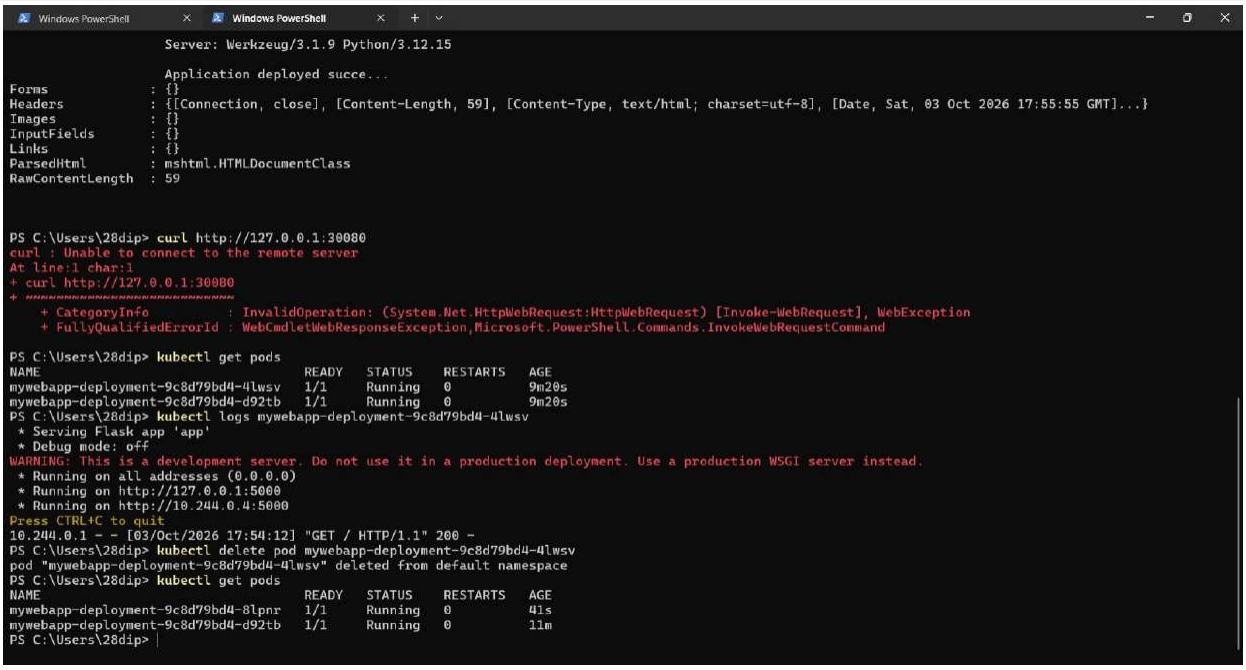
5. Application Access Output

Paste screenshot/output showing minikube service URL, browser output or curl response.



6. Scaling and Self-Healing Output

Paste screenshot/output showing kubectl scale command, increased Pod count and replacement Pod after deletion.



```
Windows PowerShell
Windows PowerShell

                    Server: Werkzeug/3.1.9 Python/3.12.15

                    Application deployed succe...
Forms             : {}
Headers           : {[Connection, close], [Content-Length, 59], [Content-Type, text/html; charset=utf-8], [Date, Sat, 03 Oct 2026 17:55:55 GMT]...}
Images            : {}
InputFields       : {}
Links             : {}
ParsedHtml        : mshtml.HTMLDocumentClass
RawContentLength  : 59


PS C:\Users\28dip> curl http://127.0.0.1:30080
curl : Unable to connect to the remote server
At line:1 char:1
+ curl http://127.0.0.1:30080
+ ~~~~~~~~~~~~~~~~~~~~~~~~~~
    + CategoryInfo          : InvalidOperation: (System.Net.HttpWebRequest:HttpWebRequest) [Invoke-WebRequest], WebException
    + FullyQualifiedErrorId : WebCmdletWebResponseException,Microsoft.PowerShell.Commands.InvokeWebRequestCommand

PS C:\Users\28dip> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
mywebapp-deployment-9c8d79bd4-4lwsv   1/1     Running   0          9m20s
mywebapp-deployment-9c8d79bd4-d92tb   1/1     Running   0          9m20s
PS C:\Users\28dip> kubectl logs mywebapp-deployment-9c8d79bd4-4lwsv
 * Serving Flask app 'app'
 * Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
 * Running on all addresses (0.0.0.0)
 * Running on http://127.0.0.1:5000
 * Running on http://10.244.0.4:5000
Press CTRL+C to quit
10.244.0.1 - - [03/Oct/2026 17:54:12] "GET / HTTP/1.1" 200 -
PS C:\Users\28dip> kubectl delete pod mywebapp-deployment-9c8d79bd4-4lwsv
pod "mywebapp-deployment-9c8d79bd4-4lwsv" deleted from default namespace
PS C:\Users\28dip> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
mywebapp-deployment-9c8d79bd4-8lpnr   1/1     Running   0          41s
mywebapp-deployment-9c8d79bd4-d92tb   1/1     Running   0          11m
PS C:\Users\28dip>
```

```
Windows PowerShell
Windows PowerShell

Press CTRL+C to quit
10.244.0.1 - - [03/Oct/2026 17:54:12] "GET / HTTP/1.1" 200 -
PS C:\Users\28dip> kubectl delete pod mywebapp-deployment-9c8d79bd4-4lwsv
pod "mywebapp-deployment-9c8d79bd4-4lwsv" deleted from default namespace
PS C:\Users\28dip> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
mywebapp-deployment-9c8d79bd4-8lpnr   1/1     Running   0          41s
mywebapp-deployment-9c8d79bd4-d92tb   1/1     Running   0          11m
PS C:\Users\28dip>
>> kubectl scale deployment mywebapp-deployment --replicas=4
deployment.apps/mywebapp-deployment scaled
PS C:\Users\28dip>
>> kubectl get pods
NAME                                  READY   STATUS    RESTARTS   AGE
mywebapp-deployment-9c8d79bd4-4fr7t   1/1     Running   0          23s
mywebapp-deployment-9c8d79bd4-8lpnr   1/1     Running   0          101s
mywebapp-deployment-9c8d79bd4-d92tb   1/1     Running   0          12m
mywebapp-deployment-9c8d79bd4-rjbsq   1/1     Running   0          23s
PS C:\Users\28dip> kubectl get deployment mywebapp-deployment
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
mywebapp-deployment   4/4     4            4           12m
PS C:\Users\28dip> kubectl get replicasets
NAME                           DESIRED   CURRENT   READY   AGE
mywebapp-deployment-9c8d79bd4   4         4         4       14m
PS C:\Users\28dip> kubectl get all
NAME                                  READY   STATUS    RESTARTS   AGE
pod/mywebapp-deployment-9c8d79bd4-4fr7t   1/1     Running   0          3m
pod/mywebapp-deployment-9c8d79bd4-8lpnr   1/1     Running   0          4m18s
pod/mywebapp-deployment-9c8d79bd4-d92tb   1/1     Running   0          15m
pod/mywebapp-deployment-9c8d79bd4-rjbsq   1/1     Running   0          3m

NAME                     TYPE        CLUSTER-IP       EXTERNAL-IP   PORT(S)        AGE
service/kubernetes       ClusterIP   10.96.0.1        <none>        443/TCP        22m
service/mywebapp-service   NodePort    10.104.140.185   <none>        80:30080/TCP   15m

NAME                                  READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/mywebapp-deployment   4/4     4            4           15m

NAME                                     DESIRED   CURRENT   READY   AGE
replicaset.apps/mywebapp-deployment-9c8d79bd4   4         4         4       15m
PS C:\Users\28dip>
```

GitHub Repository / Lab Folder Link:

https://github.com/Vighnesh279/My-project (folder: EXP-07-Kubernetes-Deployment)

Docker Hub / Container Image Link:

Local image mywebapp:1.0 (built in Experiment 6 and loaded into Minikube using minikube image load).

Final Observation Statement:

A local Minikube cluster was started and the mywebapp:1.0 image was deployed using deployment.yaml (2 replicas) and exposed through a NodePort service (service.yaml). The application was reached using the Minikube service URL, scaled from 2 to 4 replicas, and a deleted Pod was automatically replaced, showing Kubernetes self-healing.

Expected Output

A local Kubernetes cluster should start successfully. The Deployment should create the requested number of Pods. The Service should expose the application through a stable endpoint. The application should be reachable through the Service URL. Scaling should increase the number of Pods, and self-healing should recreate a Pod if one is deleted.

Conclusion

A containerized application was deployed and managed on Kubernetes. The experiment demonstrated declarative deployment using YAML manifests, service exposure, scaling of replicas and Kubernetes self-healing behavior.

Viva / Review Questions

1. What is a Pod and how does it differ from a container?
2. What is the role of a ReplicaSet within a Deployment?
3. Differentiate ClusterIP, NodePort and LoadBalancer Service types.
4. How does Kubernetes achieve self-healing?

Marks: ____ / 10 **Date: _____** **Signature of Faculty: _____**